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Assignment #1

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Algorithmic Bias in AI Systems: A Social and Ethical Analysis

As artificial intelligence systems become increasingly integrated into critical decision-making processes, the issue of algorithmic bias has emerged as one of the most pressing ethical concerns in computing technology. This topic particularly interests me because AI algorithms now influence everything from job applications and loan approvals to criminal sentencing and medical diagnoses, directly impacting millions of lives.

The core problem lies in how machine learning models, despite appearing objective, often perpetuate and amplify existing societal biases. When training data reflects historical discrimination—such as biased hiring practices or discriminatory policing—the resulting algorithms learn and reproduce these patterns. For instance, facial recognition systems have demonstrated significantly higher error rates for darker-skinned individuals, while predictive policing tools have been shown to disproportionately target minority communities.

From an ethical perspective, this raises fundamental questions about fairness, accountability, and transparency. The "black box" nature of many AI systems makes it difficult to identify when and how bias occurs, complicating efforts to ensure equitable outcomes. Moreover, the delegation of consequential decisions to automated systems challenges traditional notions of moral responsibility—when an algorithm makes a discriminatory decision, who bears the blame?

Socially, algorithmic bias threatens to deepen existing inequalities rather than alleviate them. As AI systems scale across industries, biased decisions can be propagated at unprecedented speed and scale, creating systemic disadvantages for already

marginalized groups. Addressing this issue requires not only technical solutions like fairness-aware machine learning, but also interdisciplinary collaboration involving ethicists, policymakers, and affected communities to establish appropriate governance frameworks.

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